

INITIAL PROJECT PROPOSAL

1. Project Title

Student Management System

2. Project Description

The proposed **Student Management System** is a web-based application designed to help ABC Institute manage basic student information in a centralized and organized manner.

The system will allow authorized administrators to manage student records, while students will be able to log in and view their own profile information.

The initial version will focus on essential student management functionality, with additional features considered in future releases.

3. Problem Statement

ABC Institute currently manages student information using paper records and Excel spreadsheets. This makes it difficult and time-consuming to search for information, update records, and maintain accurate student data.

The proposed system will provide a centralized web-based solution to improve the management, accessibility, and security of student information.

4. Project Objectives

The main objectives of the project are:

- To develop a web-based system for managing student information.
 - To provide secure user authentication.
 - To allow administrators to manage student records.
 - To allow students to view their own profile information.
 - To reduce manual student record management.
 - To improve the accuracy and consistency of student information.
 - To provide centralized storage of student data.
 - To demonstrate Software Change Management practices throughout the project lifecycle.
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5. Target Users

Administrator

The administrator will manage student records and access the main student management functions.

Student

Students will log in to the system and view their own profile information.

6. Main System Features

The initial Version 1.0 will provide:

Authentication

- User login
- User logout
- Secure password management

Student Management

- Add student
- View student list
- View student details
- Update student information
- Delete student records

Student Profile

- Student login
 - View own profile
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7. Proposed Technology Stack

Component	Technology
Frontend	React
Backend	Node.js with Express
Database	MySQL
Version Control	Git and GitHub
API Testing	Postman

8. Expected Outcome

The project is expected to produce a working **Version 1.0 Student Management System** that provides secure authentication and basic student record management.

The project will also establish a foundation for controlled software evolution, with future requirements managed through the Software Change Management process.